



Lec # 02
ES111

Sample Space,
Events,
Counting

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Discussion Plan

1. Sample Space

2. Events

3. Counting Sample points



Why we have *uncertainty* in The world ??



Why we have **uncertainty** in The world ??
→ lack of knowledge



Why we have uncertainty in The world ??
→ lack of knowledge

1. Future events



Why we have uncertainty in The world ??
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1. Future events



2. Too hard to collect information

Why we have uncertainty in The world ??
→ lack of knowledge

1. Future events



2. Too hard to collect information
e.g. Which places you visited today



Why we have uncertainty in The world ??
lack of knowledge

1. Future events



2. Too hard to collect information
e.g. Which places you visited today

uncertainty vanishes
↑



Why we have uncertainty in The world ??
lack of knowledge

1. Future events



2. Too hard to collect information
e.g. Which places you visited today

3. Quantum randomness

uncertainty vanishes
↑



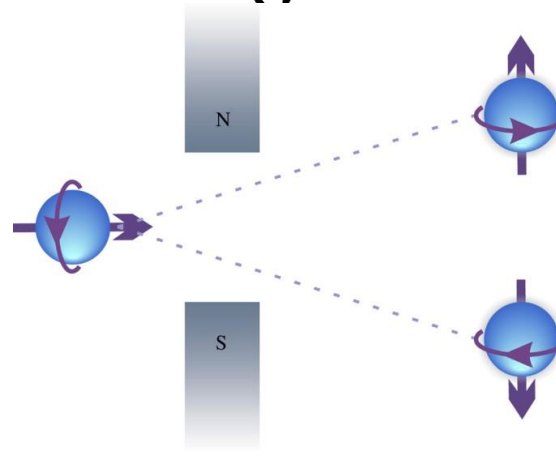
Why we have uncertainty in The world ??
lack of knowledge

1. Future events



2. Too hard to collect information
e.g. Which places you visited today

3. Quantum randomness
events are truly random
at sub-atomic scale



uncertainty vanishes



To calculate probability

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1. calculate "sample space"

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1. calculate "sample space"
2. Identify "event" of interest

To calculate probability

1. calculate "sample space"
2. Identify "event" of interest
3. Divide "event" by "sample space" $\longrightarrow$ probability

To calculate probability

1. calculate "sample space"  Today's class
2. Identify "event" of interest
3. Divide "event" by "sample space"  probability

Sample Space

Sample Space

1. Flipping a coin
2. How long it takes to fall an object from certain height
3. Rolling a die 20 times
4. Heating water to observe boiling point

Sample Space

Which statements involve uncertainty?

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Sample Space

Which statements involve uncertainty?



Flipping a coin



How long it takes to fall an object from certain height



Rolling a die 20 times



Heating water to observe boiling point

Sample Space

Which statements involve uncertainty?



Flipping a coin



How long it takes to fall an object from certain height



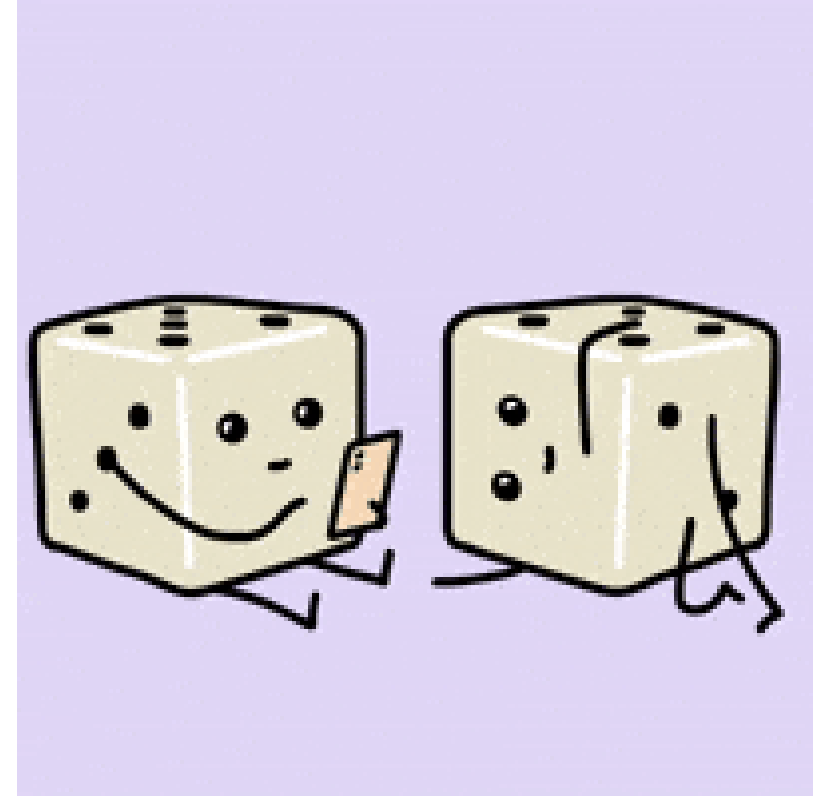
Rolling a die 20 times



Heating water to observe boiling point

A process whose outcome is uncertain, called statistical experiment.

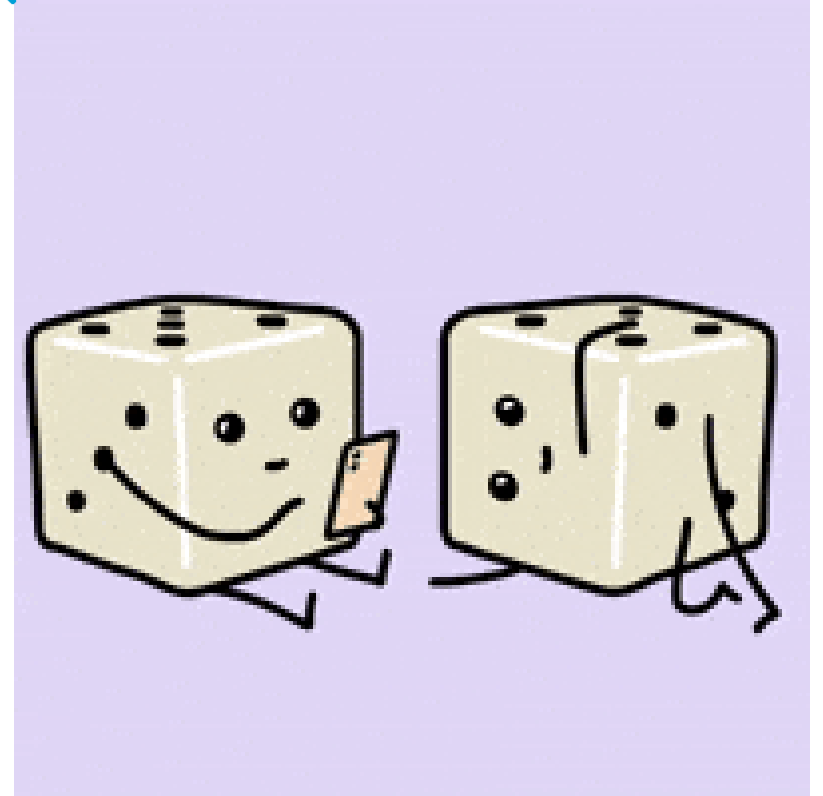
Sample Space



statistical experiment

Sample Space

A *set* of possibilities

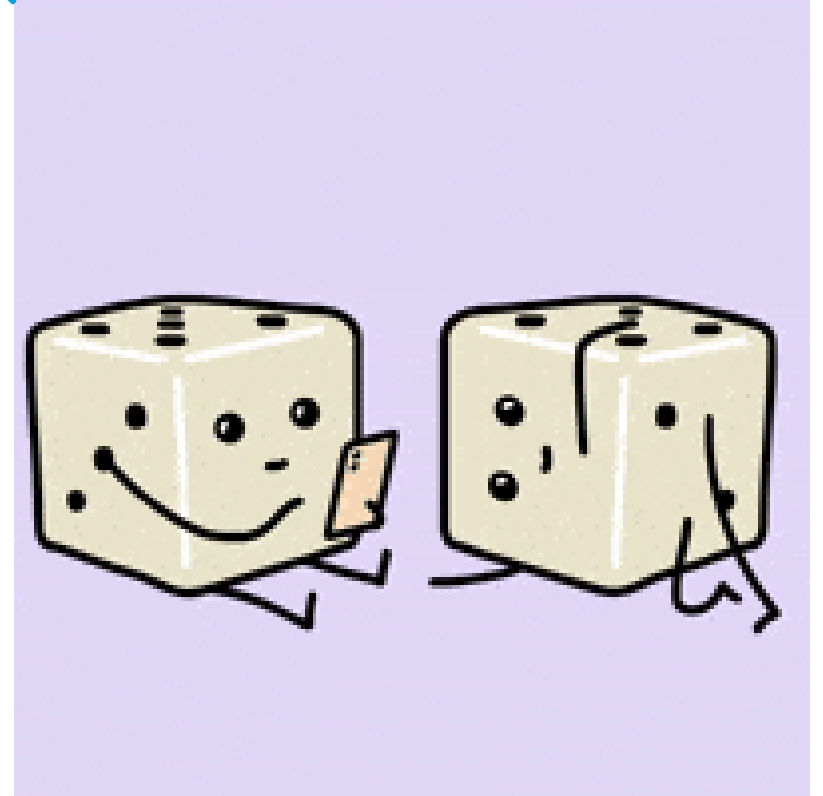


statistical experiment

Sample Space

A *set* of possibilities

Suppose we roll a *die*

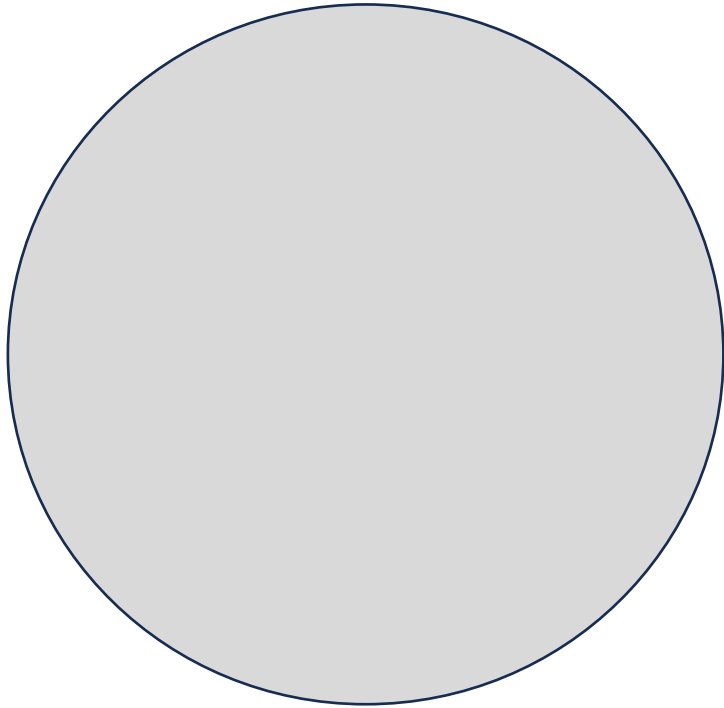


statistical experiment

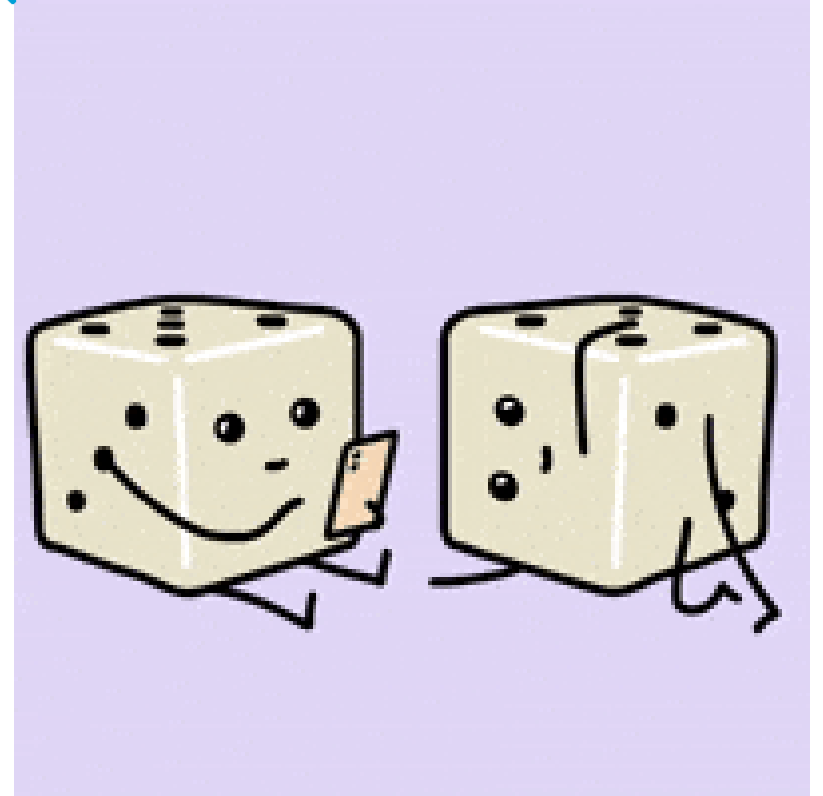
Sample Space

A *set* of possibilities

Suppose we roll a *die*



A *set* of possible outcomes

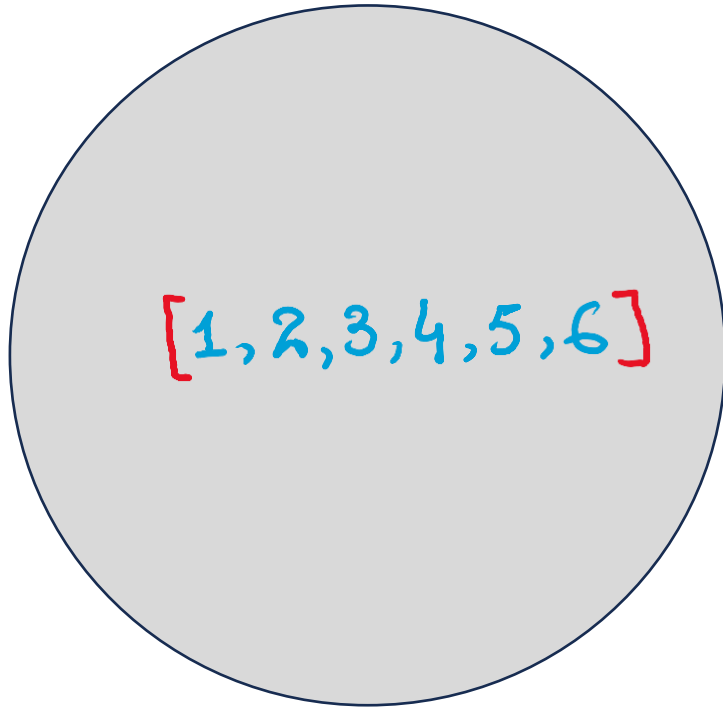


statistical experiment

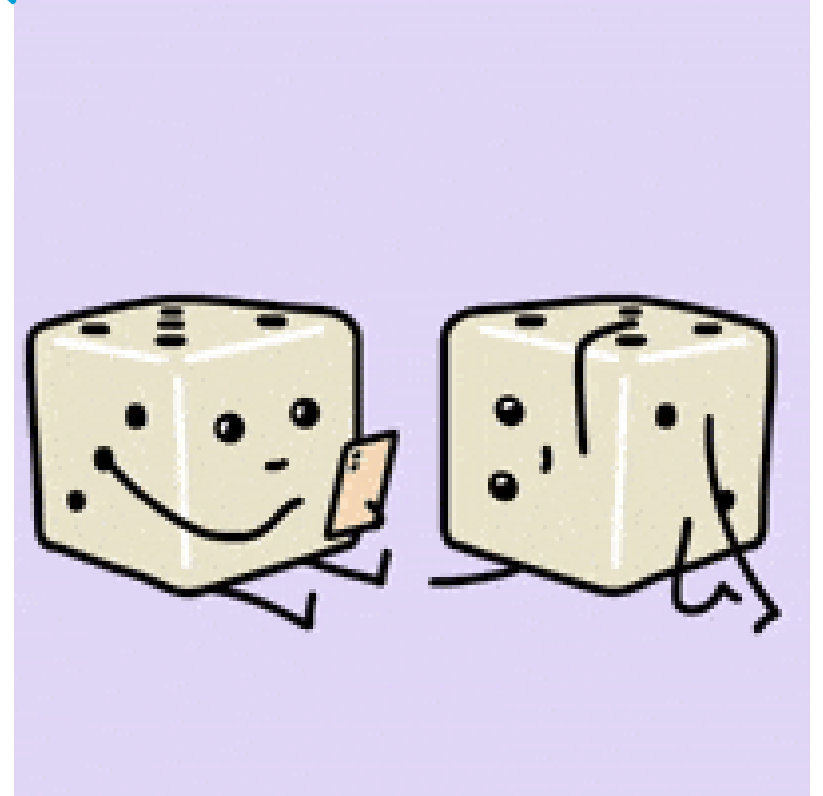
Sample Space

A *set* of possibilities

Suppose we roll a *die*



A *set* of possible outcomes



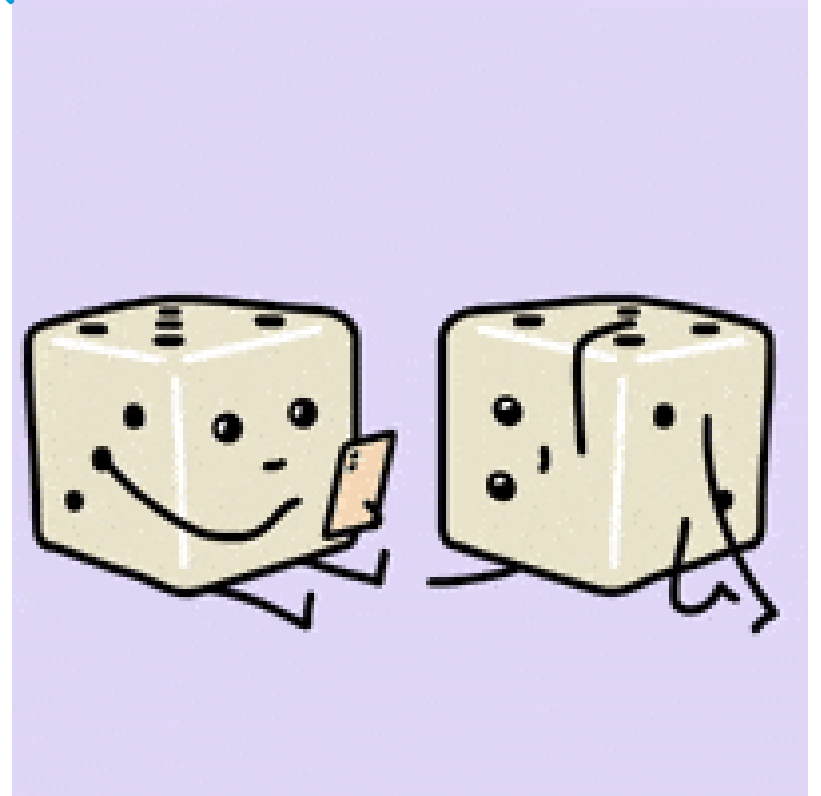
statistical experiment

Sample Space



A *set* of all possible outcomes

A *set* of possibilities



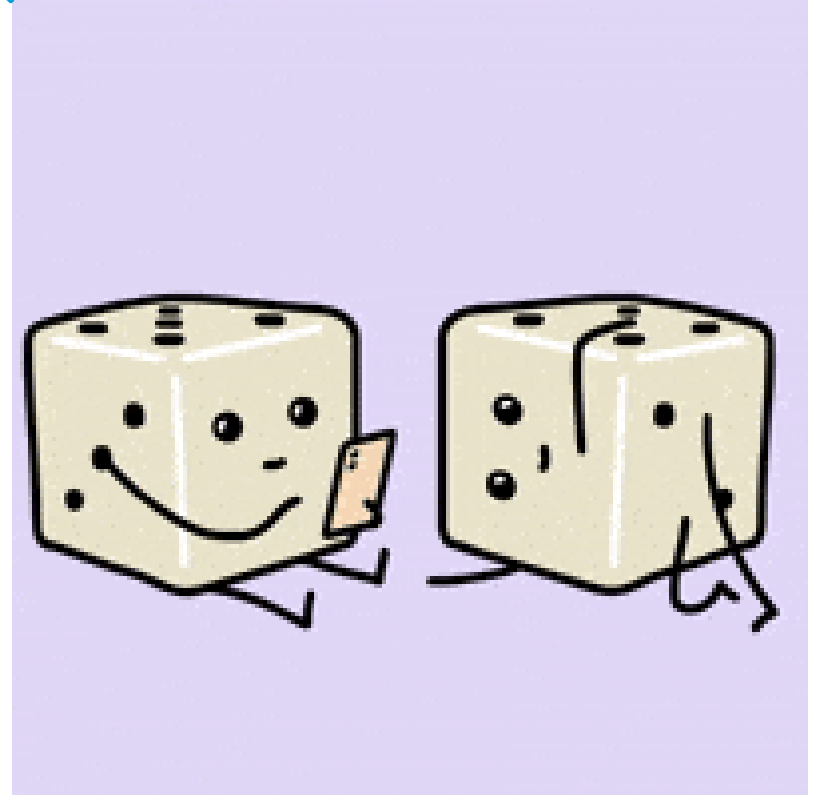
statistical experiment

Sample Space



A *set* of all possible outcomes of statistical experiment is called *sample space*

A *set* of possibilities



statistical experiment

Sample Space



A set of all possible outcomes of statistical experiment is called sample space



Consider the experiment of tossing a **coin**. Write the **sample space**.



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$$S = \{H, T\}$$



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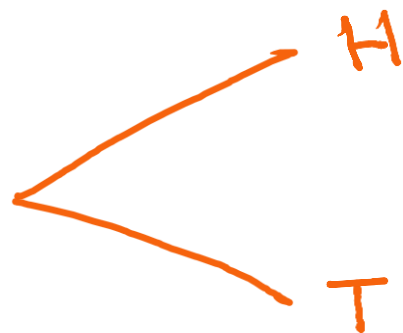
$$S = \{2, 4, 6\}$$



An experiment consists of flipping a **coin** and then flipping it a second time if a head occurs. If a tail occurs on the first flip, then a **die** is tossed once. List the **elements** of the **sample space**.

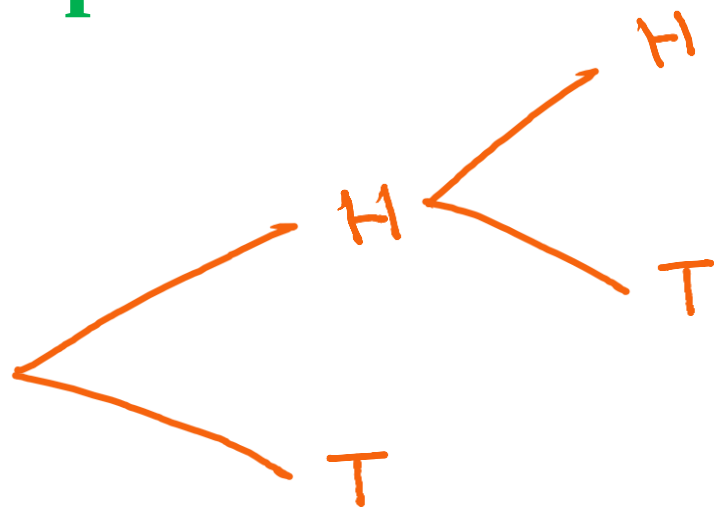


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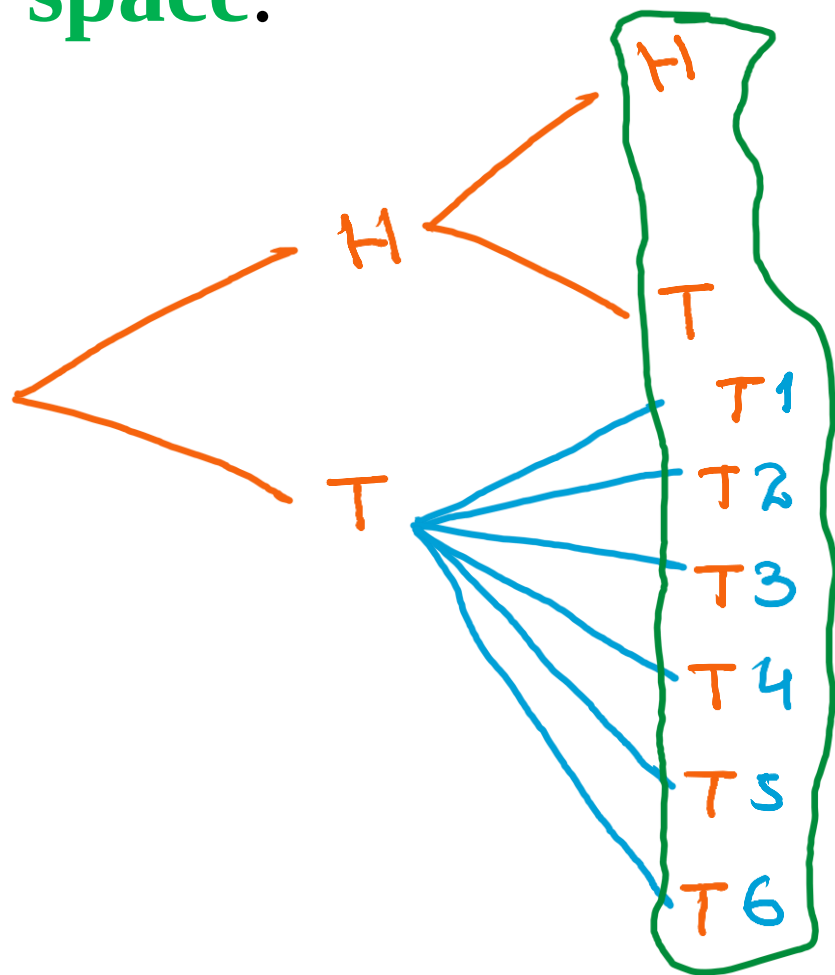


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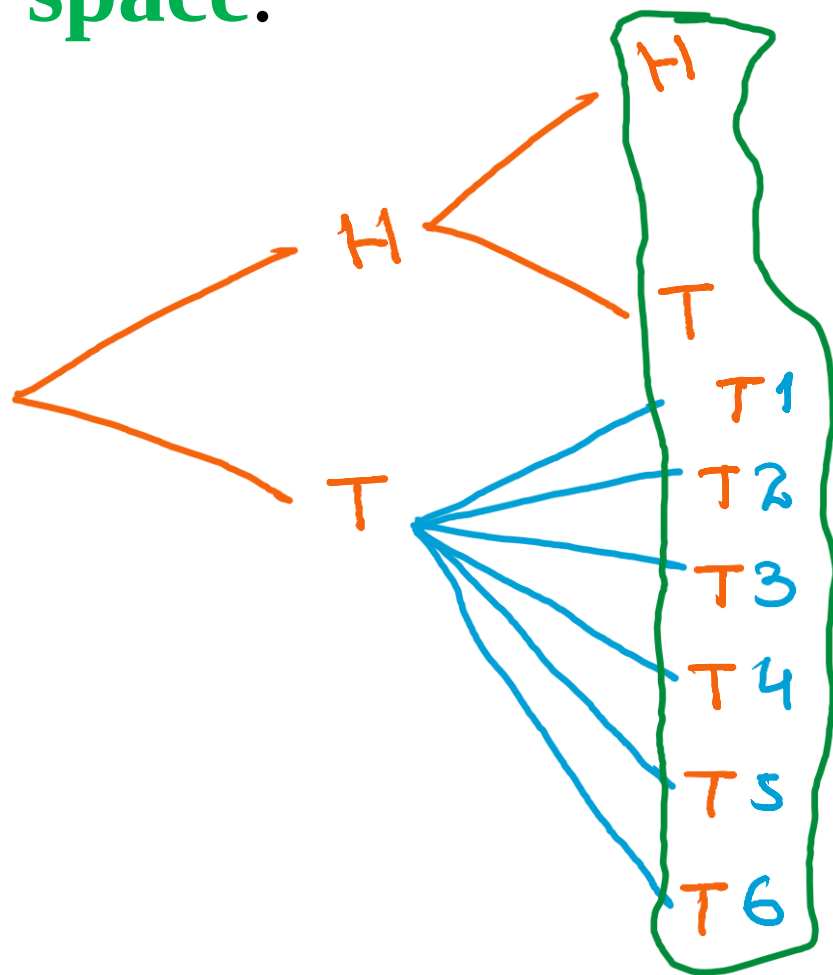


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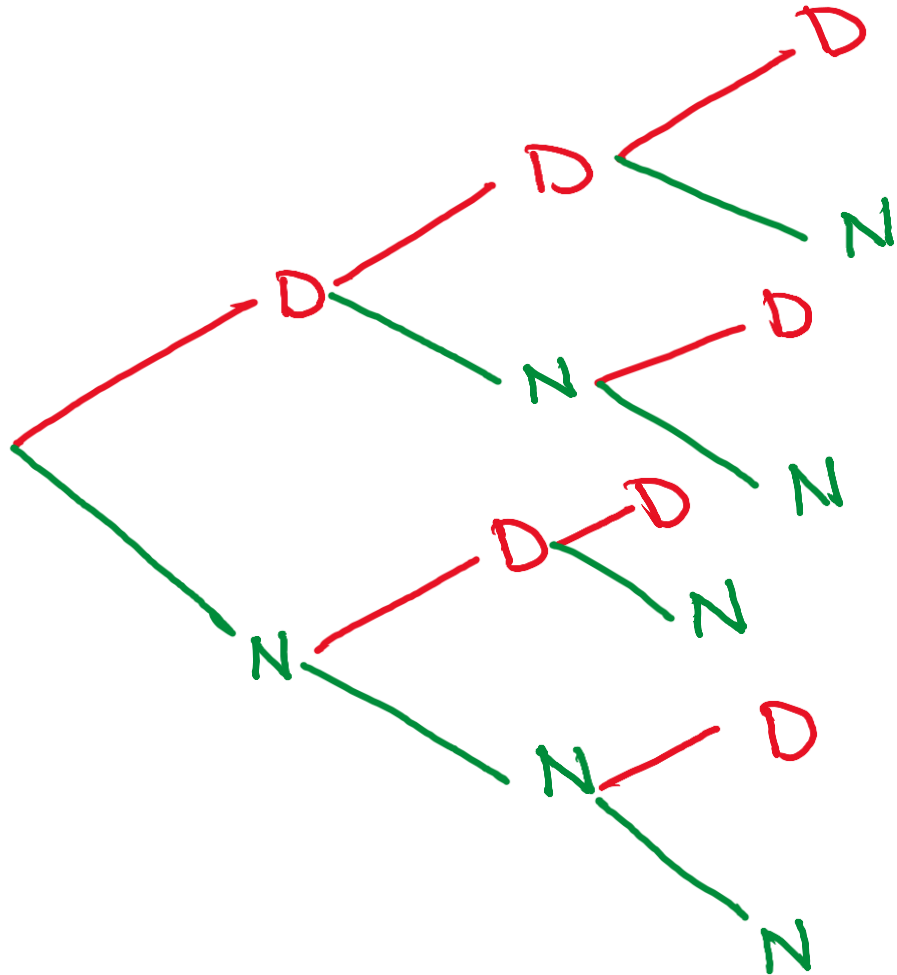
$$S = \{H, T, T1, T2, T3, T4, T5, T6\}$$



Suppose that three items are selected at random from a manufacturing process. Each item is inspected and classified **defective, D**, or **non-defective, N**. List the **elements** of **sample space**.

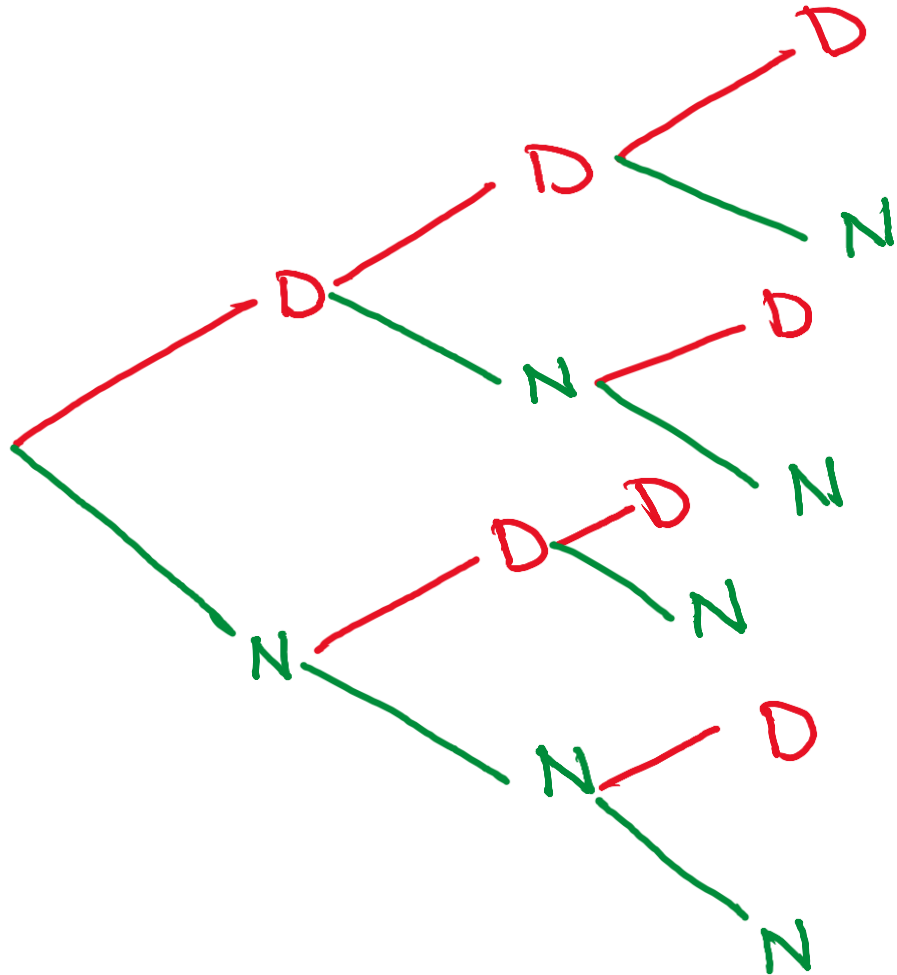


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$$S = \{ \text{DDD}, \text{DDN}, \text{DND}, \dots, \text{NNN} \}$$



If the possible outcomes of an experiment are the set of cities in the world with a population over 1 million. Write the **sample space**.



If the possible outcomes of an experiment are the set of **cities** in the world with a population over 1 million. Write the **sample space**.

$$S = \{ x \mid x \text{ is a city with a population over 1 million} \}$$



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Rule method



Use the **rule method** to describe the **sample space** S consisting of all points in the **first quadrant inside a circle of radius 3** with center at the origin.



Use the **rule method** to describe the **sample space** S consisting of all points in the **first quadrant inside a circle of radius 3** with center at the origin.

$$S = \{(x, y) \mid x^2 + y^2 \leq 9, x \geq 0, y \geq 0\}$$

Events

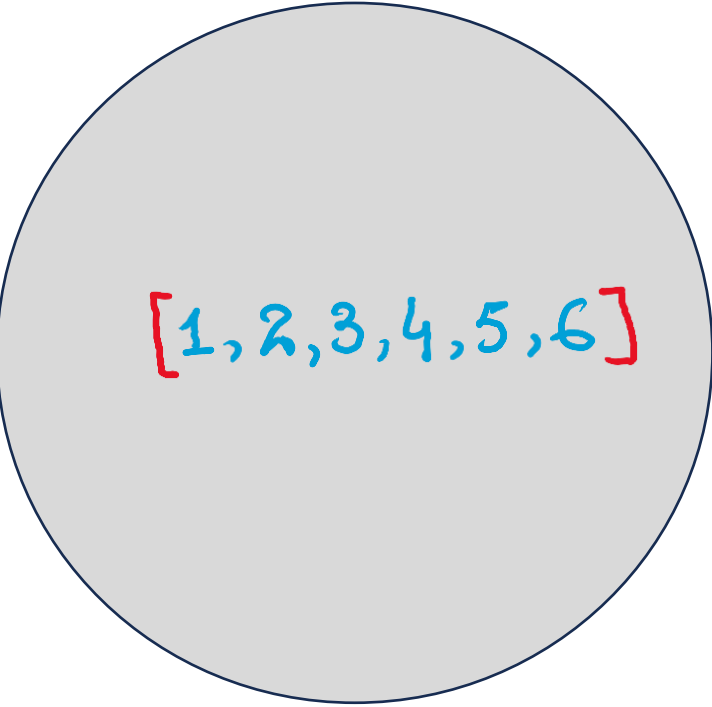
Events

An event is a subset of sample space.

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Suppose we roll a die



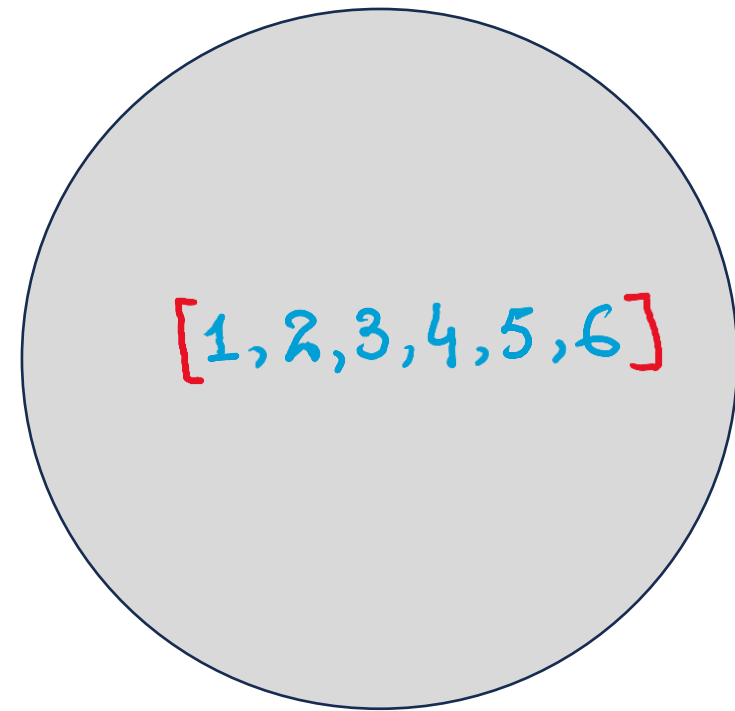
$[1, 2, 3, 4, 5, 6]$

sample space

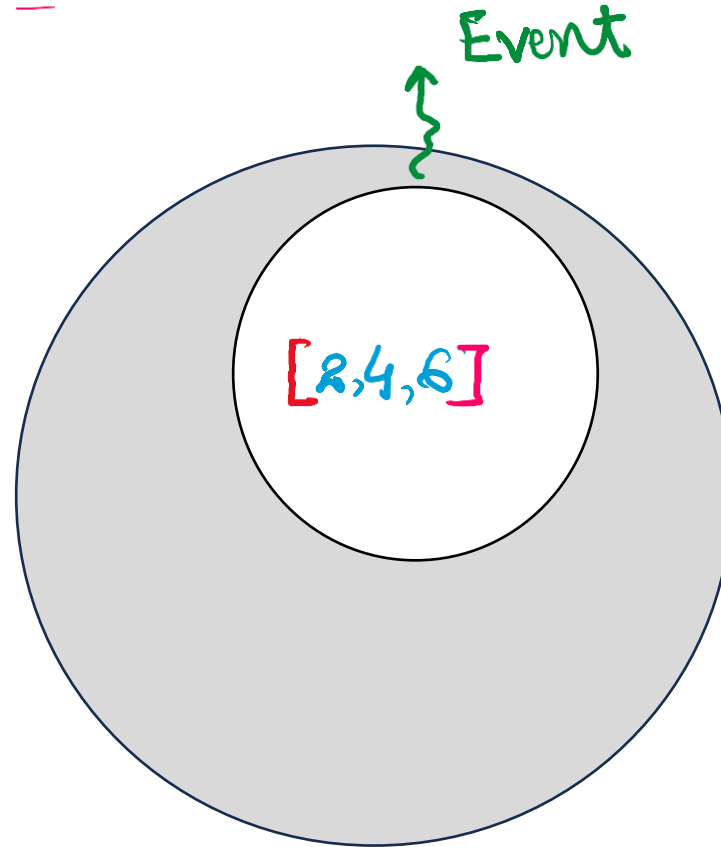
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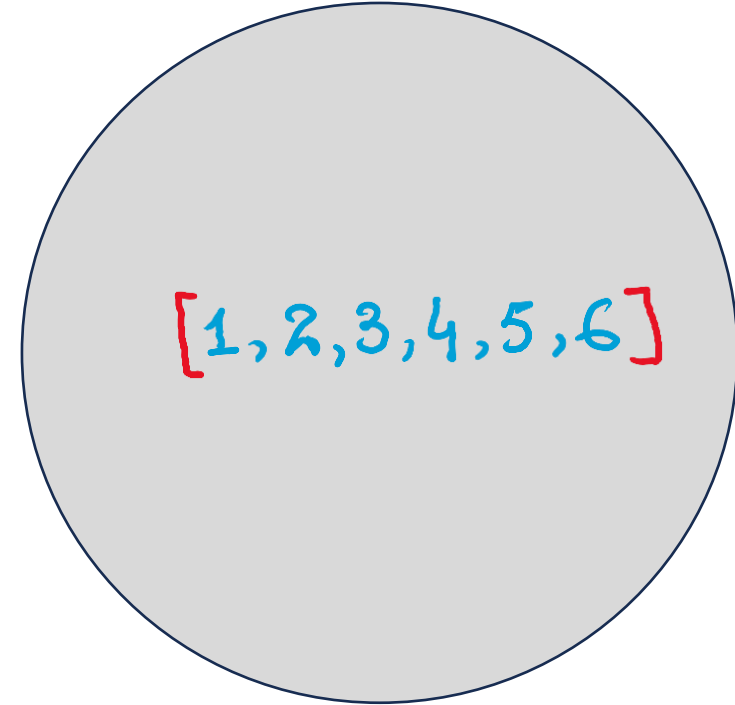
Sample space



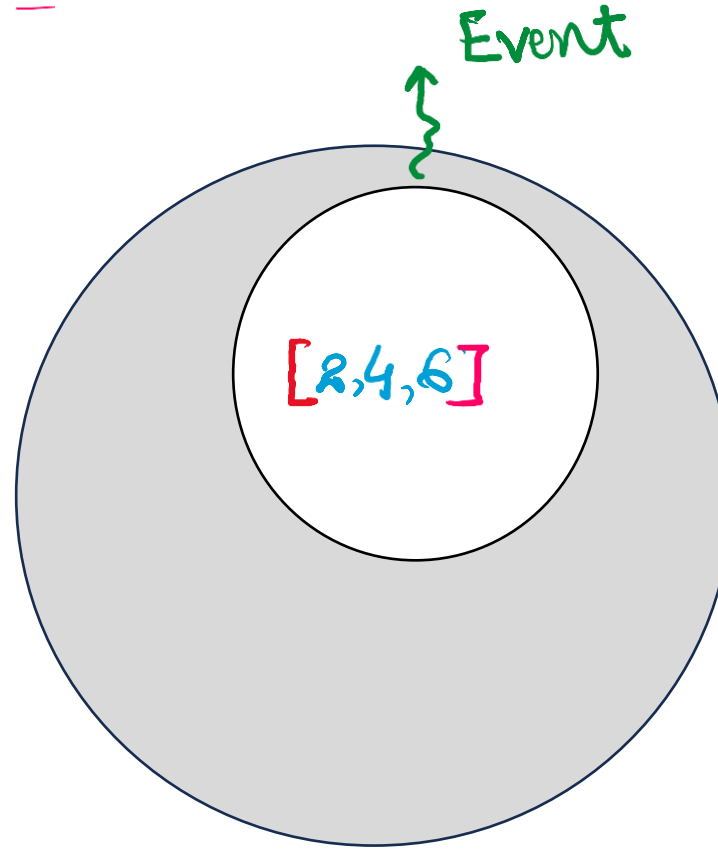
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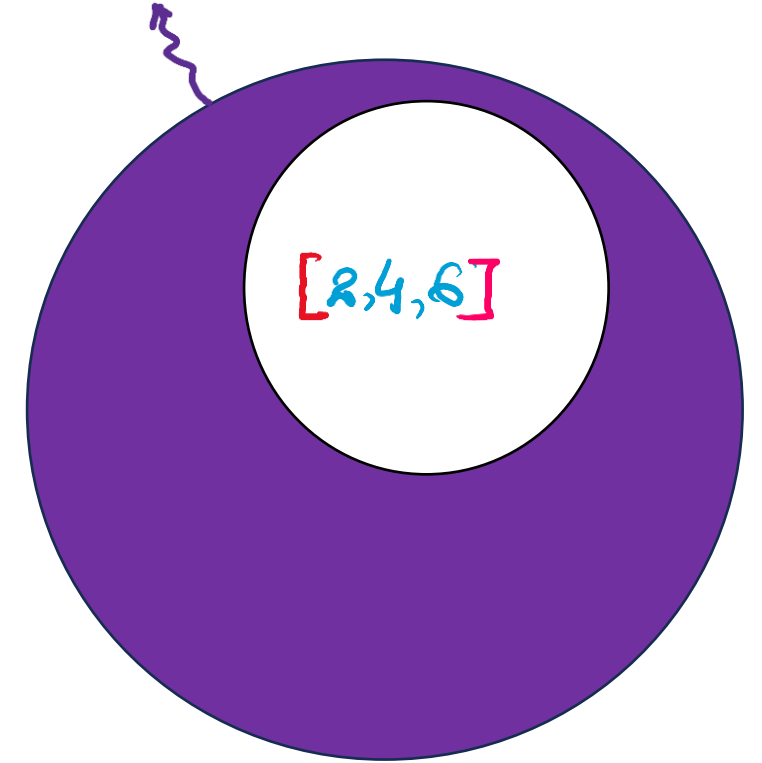


Sample space



$A = \{2, 4, 6\}$

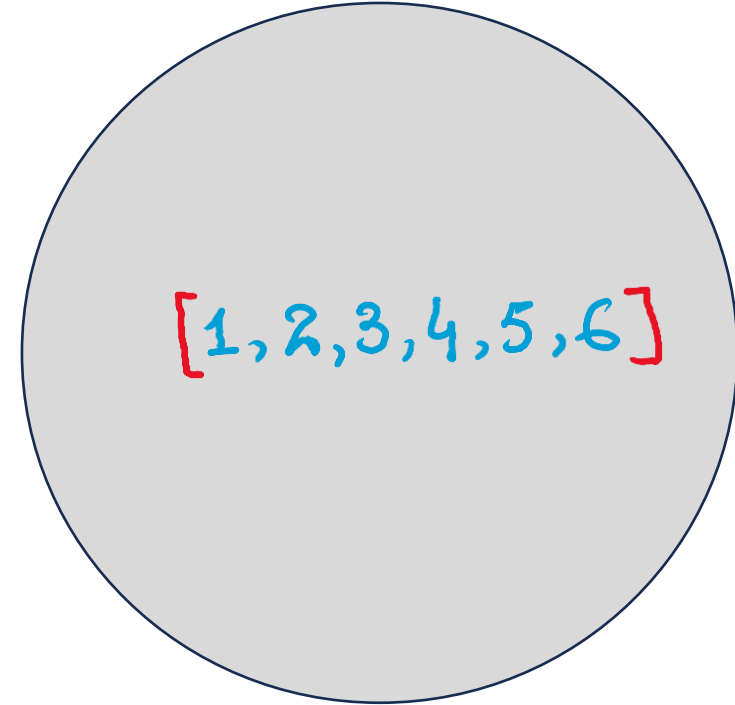
complement of Event



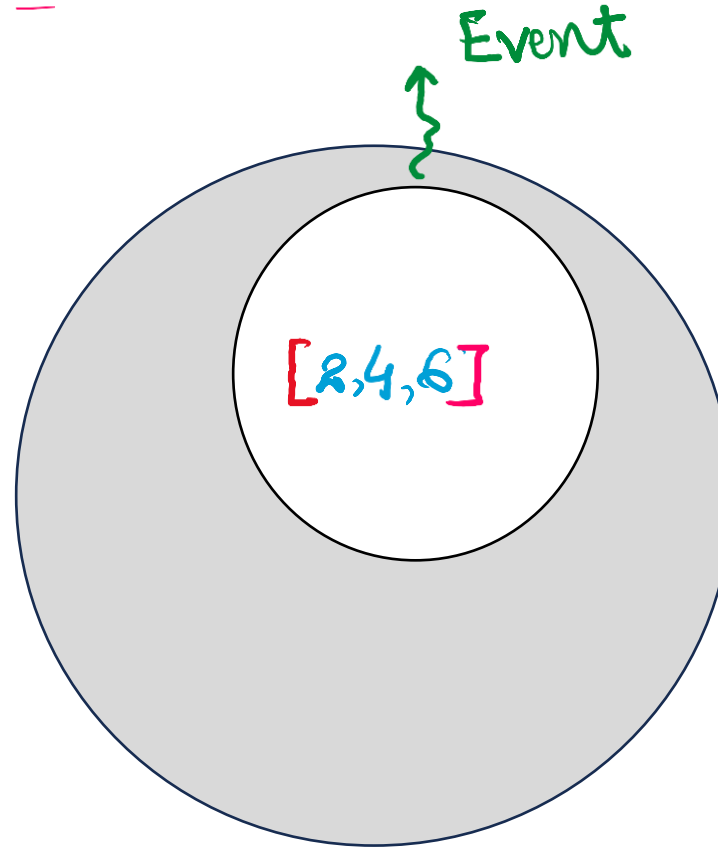
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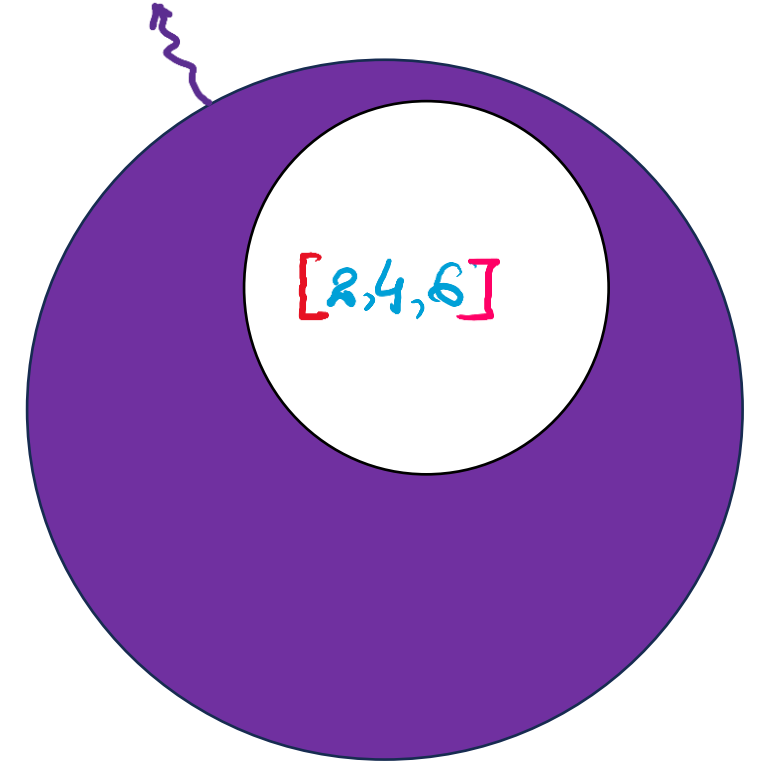


sample space



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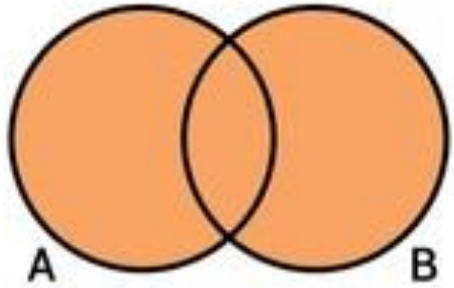
compliment of Event



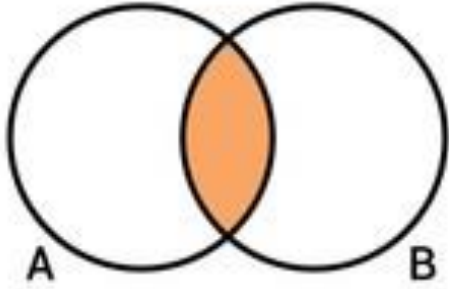
$A' = \{1, 3, 5\}$

set operations

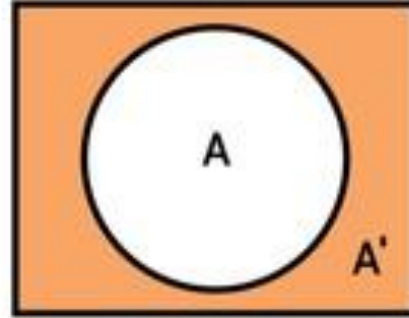
$A \cup B$



$A \cap B$



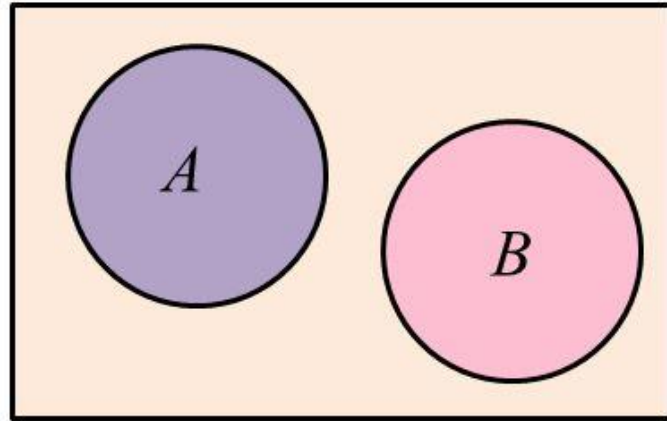
Complement of A



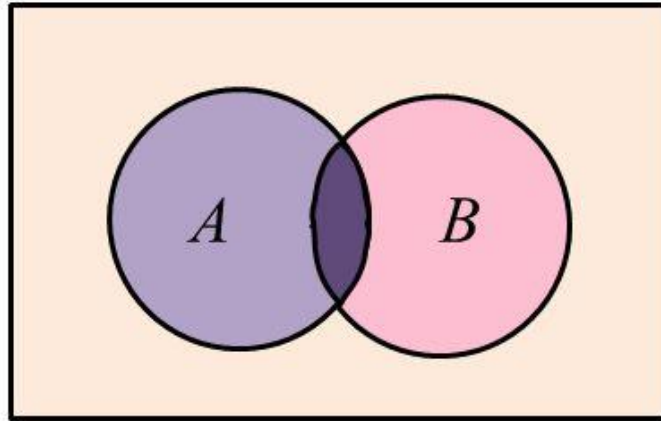
Mutually Exclusive Events

Mutually exclusive

- Two events A and B cannot occur at the same time



A and B are mutually exclusive

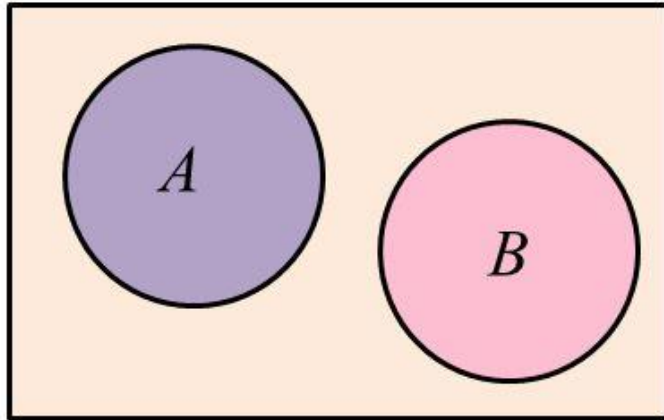


A and B are not mutually exclusive

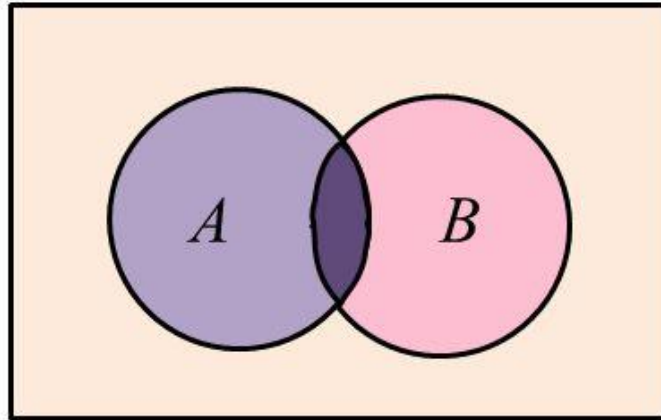
Mutually Exclusive Events

Mutually exclusive

- Two events A and B cannot occur at the same time



A and B are mutually exclusive



A and B are not mutually exclusive

$$A \cap B = \emptyset$$

Counting Sample points

Counting Sample points



2

X

1

Counting Sample points



X

1

Counting Sample points



3

X

1

Counting Sample points



3

X

Counting Sample points



3

X

2

Counting Sample points



3

X



2

X



Counting Sample points



3

X



2

X



4

Counting Sample points

Multiplication Rule



3

X



2

X



4

Counting Sample points

Multiplication Rule

categories



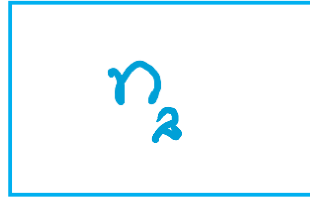
...



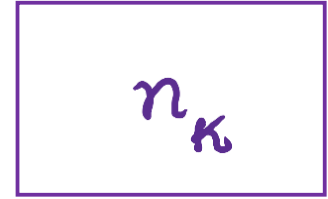
Counting Sample points

Multiplication Rule

Categories



...



All possibilities = $n_1 \times n_2 \cdots n_k$



How many sample points are there in the sample space when a pair of dice is thrown once?



How many sample points are there in the sample space when a **pair of dice** is thrown once?



$$n_1 = 6$$



$$n_2 = 6$$

$$= n_1 \times n_2 = 6 \times 6$$



Sam is going to assemble a computer by himself. He has the choice of **chips from two brands**, a **hard drive from four**, **memory from three**, and **an accessory bundle from five local stores**. How many different ways can Sam order the parts?



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$$= 2 \times 4 \times 3 \times 5 = 120$$

Ans



A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?



A certain brand of shoes comes in **5 different styles**, with each style available **in 4 distinct colors**. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

$$= 5 \times 4 = 20$$

Ans

Counting Sample points

Multiplication Rule
↓

Repetition of Events



In how many different ways can a true-false test consisting of 9 questions be answered?



In how many different ways can a **true-false** test consisting of **9 questions** be answered?



In how many different ways can a **true-false** test consisting of **9 questions** be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$



In how many different ways can a **true-false** test consisting of **9 questions** be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9$$



If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct, in how many different ways can a student check off one answer to each question?



If a multiple-choice test consists of **5 questions**, each **with 4 possible answers** of which only 1 is correct, in how many different ways can a student check off one answer to each question?

$$= 4^5$$

Ans

Counting Sample points



1. Multiplication Rule

2. Combinations

Combinations

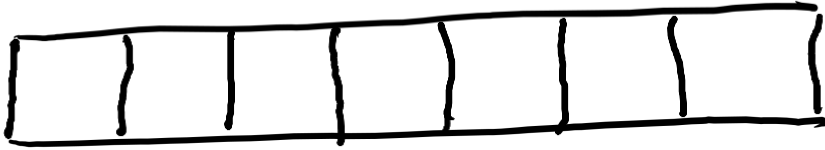
Combinations → Arrangements in which order does not matter.

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A B C same B A C

Combinations

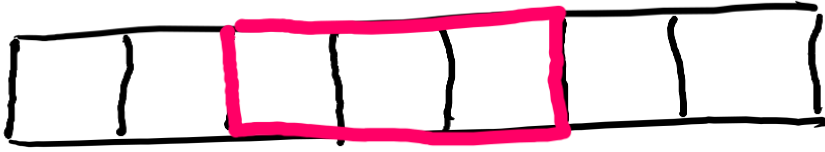
n outcomes



Combinations

γ outcomes

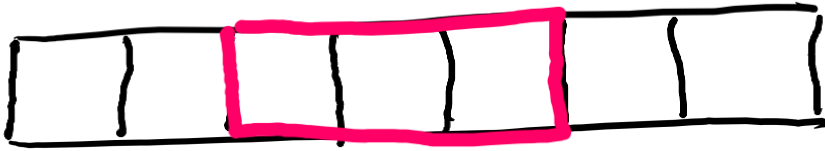
n outcomes



Combinations

r outcomes

n outcomes

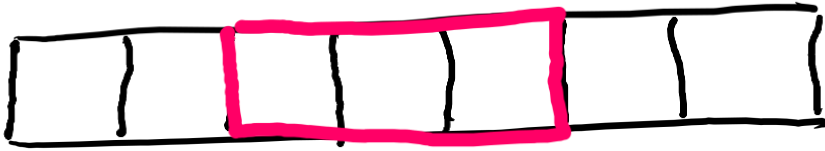


$$\text{combination} = \frac{n!}{r! (n-r)!}$$

Combinations

r outcomes

n outcomes

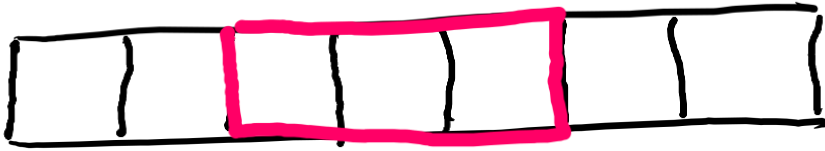


$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^n C_r$$

Combinations

r outcomes

n outcomes



$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^nC_r = \binom{n}{r}$$

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$



In one year, three same awards will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?



In one year, **three awards** (research, teaching, and service) will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?

$${}_nC_r = \frac{n!}{r! (n-r)!} = \frac{25!}{3! (25-3)!}$$



How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?